

Abhinav Bhargava

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Summary

Software Engineer with 4+ years of industry experience, specializing in backend development, distributed systems, and cloud-native applications. Skilled in designing scalable microservices, optimizing system performance, and delivering reliable, high-impact solutions.

Education

Carnegie Mellon University

August 2024 — August 2025

Master of Information Systems Management - Business Intelligence and Data Analytics

Pittsburgh, PA

• **Relevant Coursework:** Cloud Computing, Distributed Systems, Software Architectures, Machine Learning, Unstructured Data Analytics

SRM Institute of Science and Technology

July 2016 — May 2020

Bachelor of Technology in Computer Science and Engineering

Chennai, India

• **Relevant Coursework:** Algorithm Design and Analysis, Operating Systems, Computer Networks, DBMS, Data Structures

Experience

Accenture

Noida, India

Senior Software Engineer

May 2024 — July 2024

- Developed a critical infrastructure monitoring product leveraging .NET 8, engineering high-performance web APIs with layered architecture that boosted data access by 30%, supporting advanced visualizations using JavaScript and React.js.
- Engineered backend services with comprehensive observability including metrics, logging, and monitoring, integrated webhook notifications for real-time alerts, achieving 99.9% availability, and implemented CI/CD pipelines and containerization strategies.
- Executed comprehensive testing using xUnit, achieving 99.9% reliability in API performance under various load conditions.

LTIMindtree

Noida, India

Senior Software Engineer

August 2022 — May 2024

- Led cross-functional teams to design distributed systems architecture using Agile methodology, reducing delivery timelines by 30%.
- Architected enterprise-grade scalable .NET applications for Inventory Management, Material Tracking, & Incident Handling workflows applying SOLID principles and ASP.NET MVC and SQL Server database technologies, improving operational efficiency by 80%.
- Developed high-performance RESTful APIs with 99.8% uptime using a cloud-native microservices architecture with ORM integration, strategically optimized for horizontal scalability and comprehensive fault tolerance mechanisms.
- Implemented Microsoft Entra ID authentication and role-based access control (RBAC), securing applications for 10,000+ active users.
- Developed mobile applications using Xamarin Native with web API integration, boosting operational efficiency by 75% for clients.

Software Engineer

August 2020 — July 2022

- Developed a Digital Stores application for equipment marketplace, managing 50,000+ SKUs and adopted across major infra projects.
- Refactored legacy stored procedures and application logic, implementing advanced query optimization techniques, database indexing strategies, and efficient data access patterns through abstraction layers, reducing data access latency by 35%.

Projects

Twitter User Recommendation System

Java/Vert.X, Docker, Kubernetes

- Developed containerized microservices using Java, Spring Boot, Vert.X, Docker, Kubernetes, and Helm, & deployed on Amazon EKS with Terraform; integrated with Amazon RDS & benchmarked alternative managed services for cost-performance optimization.
- Engineered a scalable ETL pipeline processing 1TB+ Twitter data daily with scoring algorithm achieving 20K+ event throughput.
- Created a high-performance QR Code Service to efficiently encode/decode URL-safe Base64 strings, integrate with external authentication servers via REST/gRPC APIs, and surpassed 70K+ throughput under peak load on Kubernetes clusters.

Dynamic Resource Provisioning

Java

- Devised and implemented AWS Auto Scaling Groups with ELB and Target Groups within secure VPCs, ensuring high availability across multiple regions, handling heavy workloads efficiently, and maintaining 99% uptime through intelligent horizontal scaling.
- Automated provisioning, scaling, and security of EC2 instances through Terraform Infrastructure as Code with CloudWatch monitoring and auto-scaling policies, optimizing deployment workflows and reducing manual overhead in large-scale production environments.

Stream Processing with Kafka and Samza

Java, Kafka, Samza

- Programmed a real-time stream pipeline using Java with Kafka and Samza on Linux infrastructure to process GPS data for driver matching & ad targeting, sustaining 10K+ events/sec throughput while lowering driver match latency by 30% via efficient stream joins.
- Incorporated Test-Driven Development with 100% test coverage to ensure fault tolerance and correctness in data pipelines.

Superhero Mobile App – Distributed Systems

Java, Kotlin

- Developed a distributed superhero information system with Android mobile app, Java servlet web service, and MongoDB Atlas database, integrating third-party Superhero API to deliver real-time character data with comprehensive error handling and logging.

Skills

Languages & Operating Systems: C#, Java, Xamarin, Kotlin, Python, Scala, TypeScript, HTML, CSS, JavaScript, Linux, Windows, MacOS

Cloud & Distributed Systems: AWS (ASG, ELB, EC2, CloudWatch, EKS), Azure, GCP, Kubernetes, Apache Kafka, Docker, Terraform

Databases: RDBMS, MySQL, SQL Server, MongoDB (NoSQL), PostgreSQL, Amazon RDS, Redis, Entity Framework

Frameworks & Technologies: Spring Boot, Spring, Vert.X, Apache Spark, React.js, Node.js, ASP.NET, .NET Core, REST, gRPC

Tools & DevOps: Git (Version Control), Maven, Helm, Software Development Life Cycle (SDLC), CI/CD, JIRA, Tableau, Grafana, Agile